DSA Midterm Exam Master Cheatsheet (Java Implementation Templates)

SECTION 1: JAVA SURVIVAL BLUEPRINT (PYTHON/R TO JAVA BRIDGE)

1.1 Minimal Exam Java Class & Method Anatomy

In Java, every variable and function must live inside a `class`. The entry point for execution is `main`.

```
// Template: Minimal runnable standalone program
public class Solution {
    public static void main(String[] args) {
        // Quick variable definitions
        int n = 10;
        int[] arr = {5, 2, 8, 1, 9};

        // Calling static helper methods
        bubbleSort(arr);

        // Printing output
        System.out.println(java.util.Arrays.toString(arr));
    }

    // Method template: public static <return_type>
<name>(<params>)
    public static void bubbleSort(int[] a) {
        // Implementation here
    }
}
```

1.2 Primitives vs. References & Memory Model

- **Primitive Types**: `int`, `long`, `double`, `boolean`, `char`. Stored directly on the Call Stack. Fast, pass-by-value.
- **Reference Types**: Arrays (`int[]`), Objects (`Node`, `String`, `Stack`). Variable stores a 64-bit memory address (pointer) pointing to data on the Heap.
- **The Equality Trap (`==` vs `.equals()`)**:
  - `a == b` for primitives compares actual values (`5 == 5` is `true`).
  - `obj1 == obj2` for objects compares memory addresses (identity), NOT contents!
  - `str1.equals(str2)` compares string contents. In an exam: use `==` for `int` keys, `.equals()` for `String` objects.

1.3 Array Mechanics & Memory Contiguity

Arrays in Java are fixed-size, 0-indexed, contiguous memory blocks.

```
// 1. Allocation
int[] arr = new int[5];              // Initialized to all 0s: [0,
0, 0, 0, 0]
int[] preset = {10, 20, 30, 40};      // Size is fixed at 4
int len = preset.length;             // .length is a field, NOT a
function ()

// 2. Traversal patterns
for (int i = 0; i < arr.length; i++) { /* Standard forward scan
*/ }
for (int i = arr.length - 1; i >= 0; i--) { /* Backward scan */ }
for (int val : preset) { /* Enhanced for-each (read-only) */ }

// 3. Array Copying & Resizing
int[] copy = new int[arr.length * 2];
System.arraycopy(arr, 0, copy, 0, arr.length); // Ultra-fast O(N)
memory copy
int[] cloned = arr.clone();                    // Shallow copy

// 4. Two-Dimensional Arrays (Matrix)
int rows = 3, cols = 4;
int[][] matrix = new int[rows][cols];
int numRows = matrix.length;
int numCols = matrix[0].length;
```

1.4 Common Exam Bugs & Defensive Guard Rails

1. **`NullPointerException` (NPE)**: Attempting to access fields on `null` (e.g., `current.next` when `current == null`). Always guard: `while (current != null && current.next != null)`.
2. **`ArrayIndexOutOfBoundsException`**: Accessing `arr[-1]` or `arr[arr.length]`. Valid indices: `0 <= idx < arr.length`.
3. **Integer Division Truncation**: `5 / 2` evaluates to `2`! For floating point, cast at least one operand: `(double) 5 / 2 == 2.5`.
4. **Binary Search Midpoint Overflow**: `int mid = (low + high) / 2;` can overflow 32-bit signed int if `low + high > 2^31 - 1`. Always write: `int mid = low + (high - low) / 2;` or `int mid = (low + high) >>> 1;`.
5. **Exam Rounding Convention**: In IT013IU exams, "rounding up" convention is used: `ceil(7 / 2.0) = 4`. In Java: `(a + b - 1) / b` computes ceiling of integer division `a / b`.

1.5 Standard Java Collections Cheat Sheet (If Standard Libs Permitted)

```
// Dynamic Array: ArrayList
java.util.ArrayList<Integer> list = new java.util.ArrayList<>();
list.add(10);          // O(1) amortized
int val = list.get(0);  // O(1)
list.set(0, 99);        // O(1)
list.remove(list.size() - 1); // O(1) tail removal
int sz = list.size();

// Stack (LIFO)
java.util.Stack<Integer> stack = new java.util.Stack<>();
stack.push(10);         // O(1)
int top = stack.pop();  // O(1) removes and returns
int peek = stack.peek();// O(1) inspects without removing
boolean empty = stack.isEmpty();

// FIFO Queue (Using LinkedList)
java.util.Queue<Integer> q = new java.util.LinkedList<>();
q.add(10);             // enqueue (throws on full)
q.offer(10);           // enqueue (returns false on full)
int front = q.poll();   // dequeue (returns null if empty)
int peekQ = q.peek();   // inspect front
int qSize = q.size();

// Priority Queue (Binary Heap)
// Default: MIN-HEAP (smallest item at top)
java.util.PriorityQueue<Integer> minPQ = new
java.util.PriorityQueue<>();
minPQ.add(20); minPQ.add(5); minPQ.add(15);
int smallest = minPQ.poll(); // Returns 5

// MAX-HEAP (largest item at top)
java.util.PriorityQueue<Integer> maxPQ = new
java.util.PriorityQueue<>(java.util.Collections.reverseOrder());
maxPQ.add(20); maxPQ.add(5); maxPQ.add(15);
int largest = maxPQ.poll();  // Returns 20
```

SECTION 2: ASYMPTOTIC ANALYSIS & BIG-O DECISION MATRIX

2.1 Formal Definitions & Growth Orders

- **Big-O ($O(g(n))$)**: Asymptotic UPPER bound. $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. (Guaranteed worst-case performance).
- **Big-Omega ($\Omega(g(n))$)**: Asymptotic LOWER bound. $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$. (Best-case baseline).
- **Big-Theta ($\Theta(g(n))$)**: Asymptotic TIGHT bound. $c_1 g(n) \leq f(n) \leq c_2 g(n)$. Both upper and lower bound match.
- **Hierarchy of Growth Rates**: $O(1) < O(\log n) < O(\sqrt{n}) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$

2.2 Loop Analysis Rules

1. **Single Loop (Linear $O(n)$)**: `for (int i = 0; i < n; i += c)` $\implies O(n)$.
2. **Halving / Doubling Loop (Logarithmic $O(\log n)$)**: `for (int i = 1; i < n; i *= 2)` or `for (int i = n; i > 0; i /= 2)` $\implies O(\log n)$.
3. **Nested Loops (Quadratic $O(n^2)$)**:
   - Dependent inner loop: `for (int i = 0; i < n; i++) for (int j = i + 1; j < n; j++)`
   - Total executions: $\sum_{i=0}^{n-1}(n-1-i) = racn(n-1)2 = racn^2 - n2 \in \Theta(n^2)$.
4. **Divide and Conquer (Merge Sort $O(n \log n)$)**:
   - Recurrence: $T(n) = 2T(n/2) + O(n)$.
   - By Master Theorem: $a = 2, b = 2, d = 1 \implies \log_b a = \log_2 2 = 1 = d \implies T(n) \in \Theta(n \log n)$.

2.3 Comprehensive Data Structures & Algorithms Complexity Matrix

Data Structure / Operation	Access (Index)	Search (Value)	Insertion (Head)	Insertion (Tail)	Deletion (Head)	Deletion (Tail)
Array (Static)	$O(1)$	$O(n)$	$O(n)$ shift	$O(1)$ if room	$O(n)$ shift	$O(1)$
Dynamic Array (ArrayList)	$O(1)$	$O(n)$	$O(n)$ shift	$O(1)$ amortized	$O(n)$ shift	$O(1)$
Singly Linked List (no tail)	$O(n)$	$O(n)$	$O(1)$	$O(n)$	$O(1)$	$O(n)$
Singly Linked List (with tail)	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(n)$ need prev
Doubly Linked List (with tail)	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Stack (Array or Linked)	—	$O(n)$	$O(1)$ push	—	$O(1)$ pop	—
FIFO Queue (Circular Array/ Linked)	—	$O(n)$	—	$O(1)$ enqueue	$O(1)$ dequeue	—
Deque (Double-Ended Queue)	—	$O(n)$	$O(1)$ push-Front	$O(1)$ pushRear	$O(1)$ popFront	$O(1)$ popRear
Priority Queue (Binary Heap)	$O(1)$ peek min/max	$O(n)$	$O(\log n)$ insert	$O(\log n)$ insert	$O(\log n)$ extract	—

Sorting Algorithm	Best Time	Average Time	Worst Time	Auxiliary Space	Stable?	Comparisons (Avg)	Swaps / Shifts (Avg)
Bubble Sort	$O(n)$ flag	$O(n^2)$	$O(n^2)$	$O(1)$	**YES**	$pprox n^2 / 2$	$pprox 3n^2 / 4$ writes
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$	**NO**	Exactly $ rac{n(n-1)}{2}$	At most $n-1$ swaps
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	**YES**	$pprox n^2 / 4$	$pprox n^2 / 4$ shifts
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	**YES**	$pprox n \log_2 n$	$O(n \log n)$ copies

SECTION 3: ARRAYS, TWO-POINTERS & SLIDING WINDOWS

3.1 Prefix Sum Array (Range Query in O(1))

Used to compute subarray sum `arr[L..R]` in $O(1)$ time after $O(n)$ precomputation.

```
public class PrefixSum {
    // 1-based indexing prefix array: pref[i] = sum(arr[0..i-1])
    public static int[] buildPrefix(int[] arr) {
        int n = arr.length;
        int[] pref = new int[n + 1];
        for (int i = 0; i < n; i++) {
            pref[i + 1] = pref[i] + arr[i];
        }
        return pref;
    }

    // Sum of elements from index L to R inclusive: O(1)
    public static int queryRange(int[] pref, int L, int R) {
        return pref[R + 1] - pref[L];
    }
}
```

3.2 Difference Array (Range Updates in O(1))

Allows adding `val` to all elements in range `[L, R]` in $O(1)$ time.

```
public class DifferenceArray {
    private int[] diff;

    public DifferenceArray(int[] arr) {
        int n = arr.length;
        diff = new int[n];
        diff[0] = arr[0];
        for (int i = 1; i < n; i++) diff[i] = arr[i] - arr[i - 1];
    }

    // Add val to range [L, R] in O(1)
    public void update(int L, int R, int val) {
        diff[L] += val;
        if (R + 1 < diff.length) diff[R + 1] -= val;
    }

    // Reconstruct final array in O(n)
    public int[] getResult() {
        int[] res = new int[diff.length];
        res[0] = diff[0];
        for (int i = 1; i < diff.length; i++) res[i] = res[i - 1] +
diff[i];
        return res;
    }
}
```

3.3 Two-Pointers: Converging (Opposite Ends)

```
// Reverse array in-place: O(n) time, O(1) space
public static void reverseArray(int[] a) {
    int left = 0, right = a.length - 1;
    while (left < right) {
        int tmp = a[left];
        a[left] = a[right];
        a[right] = tmp;
        left++;
        right--;
    }
}

// Two Sum in Sorted Array: find indices where a[i] + a[j] ==
target
public static boolean twoSumSorted(int[] a, int target) {
    int left = 0, right = a.length - 1;
    while (left < right) {
        int sum = a[left] + a[right];
        if (sum == target) return true;
        else if (sum < target) left++;
        else right--;
    }
    return false;
}
```

3.4 Sliding Window: Fixed Size K

```
// Maximum sum of any contiguous subarray of fixed size K: O(n)
time
public static int maxSubarraySumFixed(int[] a, int k) {
    if (a.length < k) return -1;
    int windowSum = 0;
    for (int i = 0; i < k; i++) windowSum += a[i];
    int maxSum = windowSum;

    for (int i = k; i < a.length; i++) {
        windowSum += a[i] - a[i - k]; // Slide: add incoming,
remove outgoing
        if (windowSum > maxSum) maxSum = windowSum;
    }
    return maxSum;
}
```

3.5 Sliding Window: Variable Size

```
// Smallest subarray with sum >= target: O(n) time
public static int minSubArrayLen(int target, int[] nums) {
    int left = 0, currentSum = 0, minLen = Integer.MAX_VALUE;
    for (int right = 0; right < nums.length; right++) {
        currentSum += nums[right];
        while (currentSum >= target) {
            minLen = Math.min(minLen, right - left + 1);
            currentSum -= nums[left];
            left++;
        }
    }
    return minLen == Integer.MAX_VALUE ? 0 : minLen;
}
```

SECTION 4: SEARCHING ALGORITHMS

4.1 Linear Search (Standard & Sentinel)

```
// Standard Linear Search: O(n) worst, O(1) best
public static int linearSearch(int[] a, int target) {
    for (int i = 0; i < a.length; i++) {
        if (a[i] == target) return i;
    }
    return -1;
}
```

4.2 Standard Binary Search (Iterative & Recursive)

Invariant: Array MUST be sorted in ascending order.

```
// Iterative Binary Search: O(log n) time, O(1) space
public static int binarySearch(int[] a, int target) {
    int low = 0, high = a.length - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2; // Prevents 32-bit
integer overflow
        if (a[mid] == target) return mid;
        else if (a[mid] < target) low = mid + 1;
        else high = mid - 1;
    }
    return -1; // Not found
}

// Recursive Binary Search: O(log n) time, O(log n) call stack
space
public static int binarySearchRec(int[] a, int low, int high, int
target) {
    if (low > high) return -1;
    int mid = low + (high - low) / 2;
    if (a[mid] == target) return mid;
    if (a[mid] < target) return binarySearchRec(a, mid + 1, high,
target);
    return binarySearchRec(a, low, mid - 1, target);
}
```

4.3 Binary Search in Descending Array

```
// Descending sorted array: [99, 72, 58, 32, 17, 12]
public static int binarySearchDescending(int[] a, int target) {
    int low = 0, high = a.length - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (a[mid] == target) return mid;
        // Inverted logic: if mid element is greater, target is to
the right!
        else if (a[mid] > target) low = mid + 1;
        else high = mid - 1;
    }
    return -1;
}
```

4.4 Binary Search: First Occurrence (Lower Bound) & Last Occurrence

```
// First occurrence (Lower Bound): index of first element >=
target
public static int findFirstOccurrence(int[] a, int target) {
    int low = 0, high = a.length - 1, result = -1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (a[mid] == target) {
            result = mid;
            high = mid - 1; // Keep searching left to find earlier
match!
        } else if (a[mid] < target) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }
    return result;
```

```
}

// Last occurrence (Upper Bound): index of last element <= target
public static int findLastOccurrence(int[] a, int target) {
    int low = 0, high = a.length - 1, result = -1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (a[mid] == target) {
            result = mid;
            low = mid + 1; // Keep searching right to find later
match!
        } else if (a[mid] < target) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }
    return result;
}
```

SECTION 5: ELEMENTARY SORTING ALGORITHMS

5.1 Bubble Sort (With Early Exit & Counter Instrumentation)

Bubbles the largest unsorted element to the end of the array each pass.

```
public class BubbleSort {
    public static void sort(int[] a) {
        int n = a.length;
        boolean swapped;
        // Pass i: largest elements bubble to index (n - 1 - i)
        for (int i = 0; i < n - 1; i++) {
            swapped = false;
            for (int j = 0; j < n - 1 - i; j++) {
                if (a[j] > a[j + 1]) { // Strictly > preserves
STABILITY
                    int tmp = a[j];
                    a[j] = a[j + 1];
                    a[j + 1] = tmp;
                    swapped = true;
                }
            }
            // If no swaps occurred in an entire pass, array is
already sorted!
            if (!swapped) break; // Gives O(n) best-case time
        }
    }
}
```

5.2 Selection Sort: Standard Min-to-Front (Ascending)

Scans unsorted suffix to find minimum, swaps into boundary position i.

```
public class SelectionSort {
    public static void sortMinToFront(int[] a) {
        int n = a.length;
        for (int i = 0; i < n - 1; i++) {
            int minIdx = i;
            for (int j = i + 1; j < n; j++) {
                if (a[j] < a[minIdx]) {
                    minIdx = j;
                }
            }
            // Swap minimum into index i
            if (minIdx != i) {
                int tmp = a[i];
                a[i] = a[minIdx];
                a[minIdx] = tmp;
            }
        }
    }
}
```

5.3 Selection Sort: Variant Max-to-End (Mock Exam 2 Style)

Finds the maximum in $a[0..extend]$ and swaps it to position $ ext{end}$.

```
public static void sortMaxToEnd(int[] a) {
    int n = a.length;
    // end runs from n - 1 down to 1
    for (int end = n - 1; end > 0; end--) {
        int maxIdx = 0;
        for (int j = 1; j <= end; j++) {
            if (a[j] > a[maxIdx]) {
                maxIdx = j;
            }
        }
        // Swap max element with element at end
        int tmp = a[end];
        a[end] = a[maxIdx];
        a[maxIdx] = tmp;
    }
}
```

5.4 Insertion Sort: Standard Ascending (With Shifts)

Divides array into sorted prefix $a[0..i-1]$ and unsorted suffix. Takes `key = a[i]`, shifts larger elements right, drops `key` into slot.

```
public class InsertionSort {
    public static void sortAscending(int[] a) {
        int n = a.length;
        for (int i = 1; i < n; i++) {
            int key = a[i];
            int j = i - 1;
            // Shift elements of sorted prefix that are > key to
one position right
            while (j >= 0 && a[j] > key) {
                a[j + 1] = a[j]; // 1 memory write per shift (no 3-
step swap!)
                j--;
            }
            a[j + 1] = key; // Place key into its hole
        }
    }

    // Descending version (Mock Exam 1 Style)
    public static void sortDescending(int[] a) {
        int n = a.length;
        for (int i = 1; i < n; i++) {
            int key = a[i];
            int j = i - 1;
            // For descending: shift elements smaller than key to
the right
            while (j >= 0 && a[j] < key) {
                a[j + 1] = a[j];
                j--;
            }
            a[j + 1] = key;
        }
    }
}
```

5.5 Stability Proofs & Analysis

- **Definition**: A sorting algorithm is **stable** if elements with identical keys maintain their relative input order in the output.
- **Insertion Sort is STABLE**: The shift condition `a[j] > key` is strict. When `a[j] == key`, the condition is false; the loop terminates immediately, and `key` is placed after `a[j]`. They never cross.
- **Selection Sort is UNSTABLE**: Long-range swaps can jump an element across an identical key.
  - *Counterexample*: `[2_a, 2_b, 1]`. Min is `1` at index 2. Swap `a[0]` (2_a) with `a[2]` (1). Array becomes `[1, 2_b, 2_a]`. Relative order of 2_a and 2_b is inverted!
- **Best-Case Comparisons**:
  - Insertion Sort: Already sorted $\implies a[j] > key$ fails on first check. Exactly $n-1$ comparisons, 0 shifts $\implies \mathbf{O(n)}$.
  - Selection Sort: Non-adaptive $\implies$ outer loop runs $n-1$ times; inner loop scans to end. Always $\mathbf{\text{racn}(n-1)2 \in O(n^2)}$ comparisons.

SECTION 6: STACKS & STACK APPLICATIONS

6.1 Array-Based Stack (ArrayStack)

```
public class ArrayStack {
    private int[] data;
    private int top; // Index of current top element (-1 when
empty)
    private int capacity;

    public ArrayStack(int cap) {
        this.capacity = cap;
        this.data = new int[cap];
        this.top = -1;
    }

    public boolean isEmpty() { return top == -1; }
    public boolean isFull()  { return top == capacity - 1; }
    public int size()        { return top + 1; }

    public void push(int x) {
        if (isFull()) throw new RuntimeException("Stack
Overflow");
        data[++top] = x; // Increment top first, then assign
    }

    public int pop() {
        if (isEmpty()) throw new RuntimeException("Stack
Underflow");
        return data[top--]; // Return element, then decrement top
    }

    public int peek() {
        if (isEmpty()) throw new RuntimeException("Stack is
Empty");
        return data[top];
    }
}
```

6.2 Linked List-Based Stack (LinkedStack)

Nodes are pushed and popped at the head of a singly linked list in $O(1)$ time.

```
public class LinkedStack {
    private static class Node {
        int key;
        Node next;
        Node(int k, Node n) { this.key = k; this.next = n; }
    }
    private Node first = null;
    private int size = 0;

    public boolean isEmpty() { return first == null; }
    public int size()        { return size; }

    public void push(int x) {
        first = new Node(x, first); // New node becomes new head
        size++;
    }

    public int pop() {
        if (isEmpty()) throw new RuntimeException("Stack
Underflow");
        int val = first.key;
        first = first.next; // Advance head
        size--;
        return val;
    }

    public int peek() {
        if (isEmpty()) throw new RuntimeException("Stack is
Empty");
        return first.key;
    }
}
```

6.3 Balanced Parentheses / Brackets Matching Template

```
public static boolean isBalanced(String s) {
    java.util.Stack<Character> stack = new java.util.Stack<>();
    for (int i = 0; i < s.length(); i++) {
        char c = s.charAt(i);
        if (c == '(' || c == '[' || c == '{') {
            stack.push(c);
        } else if (c == ')' || c == ']' || c == '}') {
            if (stack.isEmpty()) return false;
            char top = stack.pop();
            if (c == ')' && top != '(') return false;
            if (c == ']' && top != '[') return false;
            if (c == '}' && top != '{') return false;
        }
    }
    return stack.isEmpty(); // If stack has unmatched opening
brackets, false
}
```

6.4 Monotonic Stack: Next Greater Element (NGE)

Finds the first element to the right that is strictly greater than `a[i]`.

```
public static int[] nextGreaterElement(int[] a) {
    int n = a.length;
    int[] result = new int[n];
    java.util.Stack<Integer> stack = new java.util.Stack<>(); //
Stores INDICES

    // Traverse from right to left
    for (int i = n - 1; i >= 0; i--) {
        // Pop elements smaller than or equal to current
        while (!stack.isEmpty() && a[stack.peek()] <= a[i]) {
            stack.pop();
        }
        result[i] = stack.isEmpty() ? -1 : a[stack.peek()];
        stack.push(i);
    }
    return result;
}
```

6.5 Stack Sorting Using Strictly ONE Auxiliary Stack (Mock Exam 2 Q5)

```
public static void sortStack(java.util.Stack<Integer> input) {
    if (input == null || input.isEmpty()) return;
    java.util.Stack<Integer> aux = new java.util.Stack<>();

    while (!input.isEmpty()) {
        int tmp = input.pop();
        // Shift elements in aux that are greater than tmp back
into input
        while (!aux.isEmpty() && aux.peek() > tmp) {
            input.push(aux.pop());
        }
        aux.push(tmp);
    }

    // aux has largest on top; pouring back to input puts SMALLEST
ON TOP!
    while (!aux.isEmpty()) {
        input.push(aux.pop());
    }
}
```

SECTION 7: QUEUES, DEQUES & PRIORITY QUEUES

7.1 Circular Array Queue (CircularQueue / Ring Buffer)

Avoids shifting elements by wrapping `front` and `rear` pointers using modulo N.

```
public class CircularQueue {
    private int[] data;
    private int front; // Points to front element
    private int rear;  // Points to rear element
    private int size;
    private int capacity;

    public CircularQueue(int cap) {
        this.capacity = cap;
        this.data = new int[cap];
        this.front = 0;
        this.rear = -1;
        this.size = 0;
    }

    public boolean isEmpty() { return size == 0; }
    public boolean isFull()  { return size == capacity; }
    public int size()        { return size; }

    public void enqueue(int x) {
        if (isFull()) throw new RuntimeException("Queue
Overflow");
        rear = (rear + 1) % capacity; // Circular wrap
        data[rear] = x;
        size++;
    }

    public int dequeue() {
        if (isEmpty()) throw new RuntimeException("Queue
Underflow");
        int val = data[front];
        front = (front + 1) % capacity; // Circular wrap
        size--;
        return val;
    }

    public int front() {
        if (isEmpty()) throw new RuntimeException("Queue is
Empty");
        return data[front];
    }
}
```

7.2 Linked List-Based Queue (LinkedQueue)

Maintains `first` (head) for $O(1)$ dequeue and `last` (tail) for $O(1)$ enqueue.

```
public class LinkedQueue {
    private static class Node {
        int key;
        Node next;
        Node(int k) { this.key = k; this.next = null; }
    }
    private Node first = null;
    private Node last = null;
    private int size = 0;

    public boolean isEmpty() { return first == null; }
    public int size()        { return size; }

    public void enqueue(int x) {
        Node newNode = new Node(x);
        if (isEmpty()) {
            first = newNode;
            last = newNode;
        } else {
            last.next = newNode;
            last = newNode;
        }
        size++;
    }

    public int dequeue() {
        if (isEmpty()) throw new RuntimeException("Queue
Underflow");
        int val = first.key;
        first = first.next;
        if (first == null) last = null; // List became empty
        size--;
        return val;
    }
}
```

7.3 Array-Based Deque (Double-Ended Queue with 4 O(1) Operations)

Required in 2023 Midterm Exam Question 5.

```
public class ArrayDeque {
    private int[] arr;
    private int front;
    private int rear;
    private int size;
    private int capacity;

    public ArrayDeque(int cap) {
        this.capacity = cap;
        this.arr = new int[cap];
        this.front = -1;
        this.rear = -1;
        this.size = 0;
    }

    public boolean isFull()  { return size == capacity; }
    public boolean isEmpty() { return size == 0; }

    // 1. Insert at Front: front moves counter-clockwise
    public void insertFront(int x) {
        if (isFull()) throw new RuntimeException("Deque
Overflow");
        if (isEmpty()) {
            front = 0; rear = 0;
        } else {
            front = (front - 1 + capacity) % capacity; // Modular
backward step
        }
        arr[front] = x;
        size++;
    }
```

```
    // 2. Insert at Rear: rear moves clockwise
    public void insertRear(int x) {
        if (isFull()) throw new RuntimeException("Deque
Overflow");
        if (isEmpty()) {
            front = 0; rear = 0;
        } else {
            rear = (rear + 1) % capacity; // Modular forward step
        }
        arr[rear] = x;
        size++;
    }

    // 3. Delete from Front: front moves clockwise
    public int deleteFront() {
        if (isEmpty()) throw new RuntimeException("Deque
Underflow");
        int val = arr[front];
        if (front == rear) { // Reset to empty when last element
removed
            front = -1; rear = -1;
        } else {
            front = (front + 1) % capacity;
        }
        size--;
        return val;
    }

    // 4. Delete from Rear: rear moves counter-clockwise
    public int deleteRear() {
        if (isEmpty()) throw new RuntimeException("Deque
Underflow");
        int val = arr[rear];
        if (front == rear) { // Reset to empty
            front = -1; rear = -1;
        } else {
            rear = (rear - 1 + capacity) % capacity;
        }
        size--;
        return val;
    }
}
```

7.4 Priority Queue & Binary Heap Indexing

Binary heaps are represented in a contiguous array without explicit pointers: - Root is at index `0`. - Parent of node i: `parent(i) = (i - 1) / 2`. - Left Child of node i: `leftChild(i) = 2 * i + 1`. - Right Child of node i: `rightChild(i) = 2 * i + 2`. - **Max Heap Invariant**: `arr[parent] >= arr[child]`. `dequeue()` extracts maximum. - **Min Heap Invariant**: `arr[parent] <= arr[child]`. `dequeue()` extracts minimum.

7.5 Queue Reversal of First K Elements (Mock Exam 1 Q5)

```
public static void reverseFirstK(java.util.Queue<Integer> q, int
k) {
    if (q == null || k <= 0 || k > q.size()) return;
    java.util.Stack<Integer> stack = new java.util.Stack<>();

    // Step 1: Dequeue first k elements and push to stack (LIFO
order)
    for (int i = 0; i < k; i++) {
        stack.push(q.poll());
    }

    // Step 2: Enqueue stack items back into queue (now reversed
at back)
    while (!stack.isEmpty()) {
        q.add(stack.pop());
    }

    // Step 3: Rotate remaining (size - k) elements to preserve
original order
    int remaining = q.size() - k;
    for (int i = 0; i < remaining; i++) {
        q.add(q.poll());
    }
}
```

SECTION 8: LINKED LISTS IN-DEPTH

8.1 Singly Linked List Implementation Template

```
public class SinglyLinkList {
    public static class Node {
        public int key;
        public Node next;
        public Node(int k) { this.key = k; this.next = null; }
    }
    public Node first = null;

    // Insert at front: O(1)
    public void insertFirst(int val) {
        Node n = new Node(val);
        n.next = first;
        first = n;
    }

    // Insert into sorted (ascending) list: O(n) (Mock Exam 1 Q3a)
    public void insertNode(int value) {
        Node newNode = new Node(value);
        if (first == null || value <= first.key) {
            newNode.next = first;
            first = newNode;
            return;
        }
        Node cur = first;
        while (cur.next != null && cur.next.key < value) {
            cur = cur.next;
        }
        newNode.next = cur.next;
        cur.next = newNode;
    }

    // Delete all nodes with given value: O(n) (Mock Exam 1 Q3b)
    public void deleteAllNodes(int value) {
        while (first != null && first.key == value) {
            first = first.next;
        }
        if (first == null) return;
        Node cur = first;
        while (cur.next != null) {
            if (cur.next.key == value) {
                cur.next = cur.next.next; // Bypass matching node
            } else if (cur.next.key > value) {
                break; // Early exit optimization if list is
sorted!
            } else {
                cur = cur.next;
            }
        }
    }

    // Move last node to front: O(n) (Mock Exam 2 Q3a)
    public void moveLastToFront() {
        if (first == null || first.next == null) return;
        Node secLast = first;
        while (secLast.next.next != null) {
            secLast = secLast.next;
        }
        Node last = secLast.next;
        secLast.next = null; // Unlink last
        last.next = first;   // Prepend to front
        first = last;        // Update head
    }

    // Remove duplicates from sorted list: O(n) (Mock Exam 2 Q3b)
    public void removeDuplicates() {
        if (first == null) return;
        Node cur = first;
        while (cur.next != null) {
            if (cur.key == cur.next.key) {
                cur.next = cur.next.next; // Bypass duplicate, DO
NOT advance cur
            } else {
                cur = cur.next;
            }
        }
    }

    // Reverse list iteratively (3-pointer pattern): O(n) time,
O(1) space
    public void reverse() {
        Node prev = null;
        Node cur = first;
        Node next = null;
        while (cur != null) {
            next = cur.next; // Save next node
            cur.next = prev; // Reverse arrow
            prev = cur;      // Advance prev
            cur = next;      // Advance cur
        }
        first = prev;        // Update head
    }

    // Find Middle Node (Slow & Fast Pointer / Tortoise & Hare):
O(n)
    public Node findMiddle() {
        if (first == null) return null;
        Node slow = first;
        Node fast = first;
        while (fast != null && fast.next != null) {
            slow = slow.next;     // 1 step
            fast = fast.next.next;  // 2 steps
        }
        return slow; // Points to middle (or second middle if
even)
    }

    // Detect Cycle (Floyd's Cycle-Finding Algorithm)
    public boolean hasCycle() {
        Node slow = first, fast = first;
        while (fast != null && fast.next != null) {
            slow = slow.next;
            fast = fast.next.next;
            if (slow == fast) return true; // Collision indicates
cycle
        }
        return false;
    }
}
```

8.2 Doubly Linked List Implementation Template

```
public class DoublyLinkList {
    public static class Node {
        public int key;
        public Node next;
        public Node previous;
        public Node(int k) { this.key = k; }
    }
    public Node first = null;
    public Node last = null;

    // Delete first node: O(1) (Real Exam 2023 Q3a)
    public void delete_first() {
        if (first == null) return;
        if (first == last) { // Single node
            first = null;
            last = null;
        } else {
            first = first.next;
            first.previous = null;
        }
    }

    // Insert at last: O(1) (Real Exam 2023 Q3b)
    public void insert_last(int key) {
        Node newNode = new Node(key);
        if (first == null) {
            first = newNode;
            last = newNode;
        } else {
            last.next = newNode;
            newNode.previous = last;
            last = newNode;
        }
    }

    // Merge two sorted doubly linked lists into a new sorted
list: O(n1 + n2) (Real Exam 2023 Q3c)
    public static DoublyLinkList merge(DoublyLinkList L1,
DoublyLinkList L2) {
        DoublyLinkList result = new DoublyLinkList();
        if (L1 == null && L2 == null) return result;
        if (L1 == null || L1.first == null) return L2;
        if (L2 == null || L2.first == null) return L1;

        Node p1 = L1.first;
        Node p2 = L2.first;

        while (p1 != null && p2 != null) {
            if (p1.key <= p2.key) {
                result.insert_last(p1.key);
                p1 = p1.next;
            } else {
                result.insert_last(p2.key);
                p2 = p2.next;
            }
        }
        while (p1 != null) {
            result.insert_last(p1.key);
            p1 = p1.next;
        }
        while (p2 != null) {
            result.insert_last(p2.key);
            p2 = p2.next;
        }
        return result;
    }
}
```

8.3 The 4x5 Linked List Complexity Matrix (Real Exam 2023 Q4)

Operation	Unsorted, Singly List	Sorted Singly List	Unsorted, Doubly List	Sorted Doubly List
Search	$O(n)$	$O(n)$	$O(n)$	$O(n)$
Insert a new value	$O(1)$ at head	$O(n)$ search to slot	$O(1)$ at head	$O(n)$ search to slot
Delete a given value	$O(n)$ search	$O(n)$ search	$O(n)$ search	$O(n)$ search
Get minimum	$O(n)$ scan	$O(1)$ head	$O(n)$ scan	$O(1)$ head
Get maximum	$O(n)$ scan	$O(n)$ ($O(1)$ w/ tail)	$O(n)$ scan	$O(1)$ tail

EXAM TRAP EXPLANATIONS: 1. **Sorted List Insert is NOT $O(1)$**: Splicing takes $O(1)$, but locating the sorted position requires scanning $O(n)$ nodes! Total: $O(n)$. 2. **Delete by Value is NOT $O(1)$**: Even in a doubly linked list where node removal is $O(1)$, you must first *search* for the node containing that value ($O(n)$). Total: $O(n)$.

SECTION 9: RECURSION & MERGE SORT

9.1 Recursion Golden Rules & Mental Models

1. **Base Case**: Must have at least one branch that returns without recurring (prevents infinite recursion / `StackOverflowError`).
2. **Progress**: Every recursive call must move strictly closer to the base case (e.g., $n-1$ or $n/2$).
3. **Call Stack**: Each recursive invocation pushes a stack frame with its own local variables. Space complexity = maximum depth of recursion tree.

9.2 Merge Sort for Arrays (Full Implementation)

```
public class MergeSort {
    public static void sort(int[] a) {
        if (a == null || a.length <= 1) return;
        int[] aux = new int[a.length];
        mergeSort(a, aux, 0, a.length - 1);
    }
    private static void mergeSort(int[] a, int[] aux, int low, int
high) {
        if (low >= high) return; // Base case: 1 element
        int mid = low + (high - low) / 2;
        mergeSort(a, aux, low, mid);      // Sort left half
        mergeSort(a, aux, mid + 1, high); // Sort right half
        merge(a, aux, low, mid, high);    // Merge two sorted
halves
    }

    private static void merge(int[] a, int[] aux, int low, int
mid, int high) {
        // Copy to auxiliary array
        for (int k = low; k <= high; k++) aux[k] = a[k];

        int i = low;      // Pointer for left half
        int j = mid + 1;  // Pointer for right half

        // Merge back into a[low..high]
        for (int k = low; k <= high; k++) {
            if (i > mid)              a[k] = aux[j++];
            else if (j > high)        a[k] = aux[i++];
            else if (aux[j] < aux[i])  a[k] = aux[j++];
            else                      a[k] = aux[i++]; // Stable:
picks left if equal!
        }
    }
}
```

SECTION 10: REAL EXAM & MOCK EXAMS DRILL BLUEPRINTS

10.1 Mock Exam 1 Full Solutions & Dry-Run Traces

Question 1: Sorting Efficiency Comparison —

- **Insertion Sort**: Best on random data among elementary sorts ($pprox n^2/4$ comparisons, $pprox n^2/4$ single shifts). Adaptive: exits early once insertion slot found.
- **Bubble Sort**: Slowest ($pprox n^2/2$ comparisons, $pprox 3n^2/4$ memory writes from 3-step swaps).
- **Selection Sort**: Non-adaptive. Always exactly $rac{n(n-1)}{2}$ comparisons. Fewest swaps ($n-1$), but comparisons dominate in RAM cache.

Question 2: Descending Insertion Sort Trace — Input: `S = [17, 72, 99, 32, 58, 70, 44, 12, 23]`

Iteration	Key	Action	Array State
Init	—	Initial	`[17, 72, 99, 32, 58, 70, 44, 12, 23]`
i = 1	**72**	Shift 17, insert 72 at 0	`[72, 17, 99, 32, 58, 70, 44, 12, 23]`
i = 2	**99**	Shift 17, 72, insert 99 at 0	`[99, 72, 17, 32, 58, 70, 44, 12, 23]`
i = 3	**32**	Shift 17, insert 32 at 2	`[99, 72, 32, 17, 58, 70, 44, 12, 23]`
i = 4	**58**	Shift 17, 32, insert 58 at 2	`[99, 72, 58, 32, 17, 70, 44, 12, 23]`
i = 5	**70**	Shift 17, 32, 58, insert 70 at 2	`[99, 72, 70, 58, 32, 17, 44, 12, 23]`
i = 6	**44**	Shift 17, 32, insert 44 at 4	`[99, 72, 70, 58, 44, 32, 17, 12, 23]`
i = 7	**12**	0 shifts, insert 12 at 7	`[99, 72, 70, 58, 44, 32, 17, 12, 23]`
i = 8	**23**	Shift 12, 17, insert 23 at 6	`[99, 72, 70, 58, 44, 32, 23, 17, 12]`

Question 4(a): Max Priority Queue Trace — Sequence: `enqueue(1), enqueue(20), dequeue(), enqueue(30), enqueue(4), dequeue(), dequeue(), enqueue(50), enqueue(0), dequeue(), enqueue(2), enqueue(6), dequeue(), dequeue(), enqueue(100), dequeue(), dequeue()` - Values returned by `dequeue()`: **`20, 30, 4, 50, 6, 2, 100, 1`** - Final Queue state: `[0]`

Question 4(b): Stack Trace — Sequence: `push(10), push(20), pop(), push(0), push(1), pop(), pop(), push(2), push(30), pop(), push(3), push(4), pop(), pop(), push(100), pop(), pop()` returned by `pop()`: **`20, 1, 0, 30, 4, 3, 100, 2`** - Final Stack state: `[10]`

10.2 Mock Exam 2 Full Solutions & Dry-Run Traces

Question 1: Stability & Best-Case Comparison —

- Stability preserves input order of duplicate keys. Insertion Sort is stable (`a[j] > key` strict). Selection Sort is unstable (swaps jump over duplicates, e.g., `[2a, 2b, 1]`).
- Best Case (Already Sorted):
  – Insertion Sort: $n-1$ comparisons, 0 shifts $\Longrightarrow \mathbf{O(n)}$.

– Selection Sort: $
rac{n(n-1)}{2}$ comparisons, 0 swaps $\implies \mathbf{O(n^2)}$.

Question 2: Selection Sort Max-to-End Trace — Input: `S = [45, 12, 85, 32, 89, 45, 21]`

Iteration	Unsorted Range	Max Found	Swap Action	Array State
Init	`A[0..6]`	—	—	`[45, 12, 85, 32, 89, 45, 21]`
1	`A[0..6]`	89 @ idx 4	Swap $A[4]$ (89) $\leftrightarrow A[6]$ (21)	`[45, 12, 85, 32, 21, 45, 89]`
2	`A[0..5]`	85 @ idx 2	Swap $A[2]$ (85) $\leftrightarrow A[5]$ (45)	`[45, 12, 45, 32, 21, 85, 89]`
3	`A[0..4]`	45 @ idx 0	Swap $A[0]$ (45) $\leftrightarrow A[4]$ (21)	`[21, 12, 45, 32, 45, 85, 89]`
4	`A[0..3]`	45 @ idx 2	Swap $A[2]$ (45) $\leftrightarrow A[3]$ (32)	`[21, 12, 32, 45, 45, 85, 89]`
5	`A[0..2]`	32 @ idx 2	Self-swap $A[2]$ (32)	`[21, 12, 32, 45, 45, 85, 89]`
6	`A[0..1]`	21 @ idx 0	Swap $A[0]$ (21) $\leftrightarrow A[1]$ (12)	`[12, 21, 32, 45, 45, 85, 89]`

Question 4(a): Min Priority Queue Trace — Sequence: `enqueue(15), enqueue(10), enqueue(20), dequeue(), enqueue(5), dequeue(), dequeue(), enqueue(30), enqueue(2), dequeue()`
Values returned by `dequeue()`: **`10, 5, 15, 2`**

Question 4(b): Circular Queue Array Trace — Capacity $N = 5$. Initial: `front = 0, rear = -1, size = 0`. Sequence: `enq(10), enq(20), enq(30), deq(), enq(40), enq(50), enq(60), deq()`

`enq(10)`: `front=0, rear=0, size=1, A=[10, null, null, null, null]` - `enq(20)`: `front=0, rear=1, size=2, A=[10, 20, null, null, null]` - `enq(30)`: `front=0, rear=2, size=3, A=[10, 20, 30, null, null]` - `deq()` $\implies$ returns 10:

`front=1, rear=2, size=2, A=[10, 20, 30, null, null]` - `enq(40)`: `front=1, rear=3, size=3, A=[10, 20, 30, 40, null]` - `enq(50)`: `front=1, rear=4, size=4, A=[10, 20, 30, 40, 50]` - `enq(60)`: `rear=(4+1)%5=0`, `front=1, rear=0, size=5, A=[60, 20, 30, 40, 50]` - `deq()` $\implies$ returns 20:

`front=(1+1)%5=2, rear=0, size=4, A=[60, 20, 30, 40, 50]` - **Final Result**: `front = 2`, `rear = 0`, `size = 4`, `A = [60, 20, 30, 40, 50]`. Active queue: `30, 40, 50, 60`.

10.3 Real Exam 2023 (IT013IU Midterm) Full Solutions

Question 1: 13-Element Selection Sort Table —

Action	0	1	2	3	4	5	6	7	8	9	10	11	12
Initial	72	4	12	53	15	2	8	16	40	21	7	87	35
Step 1 (Swap 2 @ 5 $\leftrightarrow$ 0)	**2**	4	12	53	15	**72**	8	16	40	21	7	87	35
Step 2 (Self-swap 4 @ 1)	**2**	**4**	12	53	15	72	8	16	40	21	7	87	35
Step 3 (Swap 7 @ 10 $\leftrightarrow$ 2)	**2**	**4**	**7**	53	15	72	8	16	40	21	12	87	35
Step 4 (Swap 8 @ 6 $\leftrightarrow$ 3)	**2**	**4**	**7**	**8**	15	72	53	16	40	21	12	87	35
Step 5 (Swap 12 @ 10 $\leftrightarrow$ 4)	**2**	**4**	**7**	**8**	**12**	72	53	16	40	21	15	87	35
Step 6 (Swap 15 @ 10 $\leftrightarrow$ 5)	**2**	**4**	**7**	**8**	**12**	**15**	53	16	40	21	72	87	35
Step 7 (Swap 16 @ 7 $\leftrightarrow$ 6)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	53	40	21	72	87	35
Step 8 (Swap 21 @ 9 $\leftrightarrow$ 7)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	40	53	72	87	35
Step 9 (Swap 35 @ 12 $\leftrightarrow$ 8)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	53	72	87	40
Step 10 (Swap 40 @ 12 $\leftrightarrow$ 9)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	**40**	72	87	53
Step 11 (Swap 53 @ 12 $\leftrightarrow$ 10)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	**40**	**53**	87	72
Step 12 (Swap 72 @ 12 $\leftrightarrow$ 11)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	**40**	**53**	**72**	**87**

Question 2: Queue & Stack Traces (Exam Date Cipher) —
- **2(a) FIFO Queue dequeue() sequence**: **`0, 3, 0, 4, 2, 0, 2, 3`** (Encodes date `03/04/2023`).
- **2(b) LIFO Stack pop() sequence**: **`3, 4, 0, 0, 3, 2, 1330, 2`** (Encodes date and course code `1330`).

SECTION 11: FULL MOCK EXAM 1 WALKTHROUGH & PROOFS

Mock Exam 1: Detailed Solutions & Explanations

Question 1: Sorting (20 Marks)

Problem Statement — Present the idea of the Insertion Sort algorithm for an array. For a given random data, compare and explain the runtime efficiency of Insertion Sort with that of Bubble Sort and Selection Sort.

1.1 The Core Idea of Insertion Sort —

Mechanical Explanation

Insertion Sort divides the array conceptually into two regions: 1. A **sorted prefix** on the left (initially just the first element at index `0`). 2. An **unsorted suffix** on the right (indices `1` to `n - 1`).
In each iteration i (from 1 to $n-1$), the algorithm takes the first element of the unsorted region—called the **key**—and scans backwards through the sorted prefix from right to left. Elements in the sorted prefix that are larger than the key are shifted one position to the right to make room. The moment an element smaller than or equal to the key is encountered (or the beginning of the array is reached), the scan stops, and the key is placed into the vacated position.

```
flowchart TD
    subgraph Iteration_Step["Insertion Sort Step at Index i"]
        A["Current Key = A[i]"] --> B["j = i - 1"]
        B --> C{"j >= 0 and A[j] > Key?"}
        C -- Yes --> D["Shift A[j] to A[j + 1]<br/>j = j - 1"]
        D --> C
        C -- No --> E["Insert Key at A[j + 1]"]
    end
```

Bioinformatics Mental Model

Think of Insertion Sort like assembling a physical genetic map or organizing sequencing reads manually by chromosome start position: - You already have an ordered stack of mapped reads on your desk. - You pick up a new read card (the `key`). - You flip backward through your sorted stack until you find where its coordinate fits, slide the higher-coordinate cards over, and drop the new card into place.

1.2 Comparison on Random Data: Insertion Sort vs. Bubble Sort vs. Selection Sort —

All three algorithms have an asymptotic worst-case and average-case time complexity of $\mathcal{O}(n^2)$. However, on **randomly permuted data**, their real-world runtime efficiency differs significantly due to the constant factors hidden inside the $\mathcal{O}$ notation: the number of comparisons and the cost of memory writes (swaps vs. shifts).

Algorithm	Average Comparisons	Average Data Movements (Writes)	Primary Mechanism	Practical Speed on Random Data
Insertion Sort	$\approx \frac{n^2}{4}$ comparisons	$\approx \frac{n^2}{4}$ single shifts	Adaptive scan with early exit	**Fastest** among the three
Bubble Sort	$\approx \frac{n^2}{2}$ comparisons	$\approx \frac{3n^2}{4}$ memory writes ($\frac{n^2}{4}$ full swaps)	Repeated adjacent pairwise swaps	**Slowest** (heavy overhead)
Selection Sort	Exactly $\frac{n(n-1)}{2} \approx \frac{n^2}{2}$	Exactly $n-1$ swaps ($3(n-1)$ writes)	Blind scan for global minimum	Moderate (fewer writes, but maximum comparisons)

Detailed Comparison Breakdown

1. **Insertion Sort vs. Selection Sort**:
   - **Comparisons**: Selection Sort is non-adaptive. To find the minimum in the remaining unsorted subarray, it unconditionally compares every single remaining pair. For an array of size n, it **always** performs $\frac{n(n-1)}{2} \approx 0.5n^2$ comparisons, regardless of whether the array is sorted, reversed, or random. In contrast, for random data, Insertion Sort finds the correct insertion slot on average halfway through the sorted prefix. Thus, Insertion Sort performs only $\approx \frac{n^2}{4} = 0.25n^2$ comparisons—roughly **half as many comparisons as Selection Sort**.
   - **Data Movements**: Selection Sort performs at most $n-1$ swaps (which is optimal in write-sensitive media like EEPROM). However, in standard RAM cache architectures, comparisons and branching dominate for small to medium arrays, making Insertion Sort faster overall on random data.
2. **Insertion Sort vs. Bubble Sort**:
   - **Comparisons**: Standard Bubble Sort scans adjacent pairs across the entire array pass-by-pass, requiring $\approx \frac{n^2}{2}$ comparisons on random data (twice that of Insertion Sort).
   - **Swaps vs. Shifts**: When Bubble Sort detects an inversion, it performs a complete 3-step swap using a temporary variable:
     ```
     int temp = A[j]; A[j] = A[j + 1]; A[j + 1] = temp; // 3 memory writes per inversion!
     ```
     Insertion Sort simply shifts the larger element one slot to the right (`A[j + 1] = A[j];` — 1 write). Only after all shifting is complete does it write the `key` once. Because random data contains roughly $\frac{n^2}{4}$ inversions, Bubble Sort does $\frac{3n^2}{4}$ memory writes, whereas Insertion Sort does only $\approx \frac{n^2}{4}$ writes. This makes Bubble Sort drastically slower in CPU cache performance.

Question 2: Sorting (15 Marks)

Problem Statement — Given the sequence S:
$$17, 72, 99, 32, 58, 70, 44, 12, 23$$
Provide the detailed steps to arrange S in **descending order** using the Insertion Sort algorithm, showing the changes in S at each iteration.

Step-by-Step Execution —
- **Input Array Size**: $n = 9$
- **Target Order**: Descending (largest to smallest, so $A[j] < \text{key}$ causes elements to shift right).
- **Initial State**:
  $$[17, 72, 99, 32, 58, 70, 44, 12, 23]$$
  Sorted prefix initially contains index `0`: `[17]`.

Iteration	Key	Comparisons & Shifts	Array State After Iteration
Initial	—	Unsorted input	`[17, 72, 99, 32, 58, 70, 44, 12, 23]`
$i = 1$	**72**	$17 < 72 \implies$ shift 17 to right; insert 72 at index 0	`[72, 17, 99, 32, 58, 70, 44, 12, 23]`
$i = 2$	**99**	$17 < 99 \implies$ shift 17; $72 < 99 \implies$ shift 72; insert 99 at index 0	`[99, 72, 17, 32, 58, 70, 44, 12, 23]`
$i = 3$	**32**	$17 < 32 \implies$ shift 17; $72 > 32 \implies$ stop; insert 32 at index 2	`[99, 72, 32, 17, 58, 70, 44, 12, 23]`
$i = 4$	**58**	$17 < 58 \implies$ shift 17; $32 < 58 \implies$ shift 32; $72 > 58 \implies$ stop; insert 58 at index 2	`[99, 72, 58, 32, 17, 70, 44, 12, 23]`
$i = 5$	**70**	Shift 17, 32, 58; $72 > 70 \implies$ stop; insert 70 at index 2	`[99, 72, 70, 58, 32, 17, 44, 12, 23]`
$i = 6$	**44**	Shift 17, 32; $58 > 44 \implies$ stop; insert 44 at index 4	`[99, 72, 70, 58, 44, 32, 17, 12, 23]`
$i = 7$	**12**	$17 > 12 \implies$ stop immediately (0 shifts); insert 12 at index 7	`[99, 72, 70, 58, 44, 32, 17, 12, 23]`
$i = 8$	**23**	Shift 12, 17; $32 > 23 \implies$ stop; insert 23 at index 6	`[99, 72, 70, 58, 44, 32, 23, 17, 12]`

Final Result —
$$[99, 72, 70, 58, 44, 32, 23, 17, 12]$$

Question 3: Linked List (35 Marks)

Problem Statement — Given the sorted (ascending order) `SinglyLinkList` class below:

```
public class SinglyLinkList {
    private Node first;

    public void insertNode(int value) {
        // YOUR CODE
    }

    public void deleteAllNodes(int value) {
        // YOUR CODE
    }
}

public class Node {
    private int key;
    private Node next;
}
```

- **a)** (15 marks) Complete the `insertNode` method that inserts a `Node` with a given integer into the sorted list at the right location.
- **b)** (20 marks) Complete the `deleteAllNodes` method that deletes all occurrences of a given integer value from the list.

Solution 3(a): `insertNode(int value)` —

Algorithm & Edge Case Walkthrough

Because the list is maintained in **ascending order**: 1. **Case 1 (Empty list or Insert at Front)**: If `first == null` or the new `value <= first.key`, the new node becomes the new head of the list. Its `next` points to the old `first`, and `first` is updated. 2. **Case 2 (Insert Middle or End)**: Traverse the list using a pointer `current` starting at `first`. We advance while `current.next != null` and `current.next.key < value`. Once the loop terminates, `current` points to the exact node after which the new node must be spliced:
`java  newNode.next = current.next;  current.next = newNode;`

Java Implementation

```
public void insertNode(int value) {
    Node newNode = new Node();
    newNode.key = value;
    newNode.next = null;

    // Edge Case: List is empty OR value belongs before the first
```

```
node
    if (first == null || value <= first.key) {
        newNode.next = first;
        first = newNode;
        return;
    }

    // Traverse to locate the node immediately preceding the
insertion point
    Node current = first;
    while (current.next != null && current.next.key < value) {
        current = current.next;
    }

    // Splice the new node into the list
    newNode.next = current.next;
    current.next = newNode;
}
```

- **Time Complexity**: $\mathcal{O}(n)$ in the worst case (inserting at the end of the list), $\mathcal{O}(1)$ in the best case (inserting before the head).
- **Space Complexity**: $\mathcal{O}(1)$ auxiliary space.

Solution 3(b): `deleteAllNodes(int value)` —

Algorithm & Edge Case Walkthrough

Deletions in a singly linked list require updating the `next` reference of the *preceding* node: 1. **Head Cleanup**: The target `value` might appear at the head of the list (one or multiple times, e.g., `20 -> 20 -> 30`). We must advance `first = first.next` until `first == null` or `first.key != value`. 2. **Body Cleanup**: After trimming the head, if the list is empty, we stop. Otherwise, we traverse using `current = first`. - If `current.next.key == value`, we bypass that node: `current.next = current.next.next;` (we do *not* advance `current` yet, because the newly linked node might also contain `value`). - Because the list is sorted in ascending order, if `current.next.key > value`, no further occurrences of `value` can exist downstream, so we can terminate early! - Otherwise, advance `current = current.next`.

Java Implementation

```
public void deleteAllNodes(int value) {
    // Step 1: Remove all matching occurrences at the head of the
list
    while (first != null && first.key == value) {
        first = first.next;
    }

    // If the list is now empty, nothing left to delete
    if (first == null) {
        return;
    }

    // Step 2: Remove matching occurrences in the remainder of the
list
    Node current = first;
    while (current.next != null) {
        if (current.next.key == value) {
            // Bypass the target node (unlinks it from the chain)
            current.next = current.next.next;
        } else if (current.next.key > value) {
            // Optimization: since the list is sorted, we can
terminate early
            break;
        } else {
            current = current.next;
        }
    }
}
```

- **Time Complexity**: $\mathcal{O}(n)$ single pass over the list.
- **Space Complexity**: $\mathcal{O}(1)$ auxiliary memory.

Question 4: Queue and Stack (20 Marks)

4(a) Max Priority Queue Operations (10 Marks) —

Problem Statement

Show the queue after each operation of the following sequence of **MAX PRIORITY QUEUE** operations, if executed on an initially empty queue:

```
enqueue(1), enqueue(20), dequeue(), enqueue(30), enqueue(4),
dequeue(), dequeue(),
enqueue(50), enqueue(0), dequeue(), enqueue(2), enqueue(6),
dequeue(), dequeue(),
enqueue(100), dequeue(), dequeue()
```

Behavioral Rules of Max Priority Queue

- `enqueue(x)` : Adds item x into the priority queue.
- `dequeue()` : Extracts and returns the item with the **maximum numerical value**.
- Mental Model: Like an automated variant caller queue that always pulls the genetic variant with the highest quality/confidence score first.

Step-by-Step Trace

Step	Operation	Action & Priority Queue State (Ordered Highest to Lowest)	Value Returned
1	`enqueue(1)`	`[1]`	—
2	`enqueue(20)`	`[20, 1]`	—
3	`dequeue()`	Remove max element (20) ⟹ Queue: `[1]`	**20**
4	`enqueue(30)`	`[30, 1]`	—
5	`enqueue(4)`	`[30, 4, 1]`	—
6	`dequeue()`	Remove max element (30) ⟹ Queue: `[4, 1]`	**30**
7	`dequeue()`	Remove max element (4) ⟹ Queue: `[1]`	**4**
8	`enqueue(50)`	`[50, 1]`	—
9	`enqueue(0)`	`[50, 1, 0]`	—
10	`dequeue()`	Remove max element (50) ⟹ Queue: `[1, 0]`	**50**
11	`enqueue(2)`	`[2, 1, 0]`	—
12	`enqueue(6)`	`[6, 2, 1, 0]`	—
13	`dequeue()`	Remove max element (6) ⟹ Queue: `[2, 1, 0]`	**6**
14	`dequeue()`	Remove max element (2) ⟹ Queue: `[1, 0]`	**2**
15	`enqueue(100)`	`[100, 1, 0]`	—
16	`dequeue()`	Remove max element (100) ⟹ Queue: `[1, 0]`	**100**
17	`dequeue()`	Remove max element (1) ⟹ Queue: `[0]`	**1**

- **Final Queue State**: `[0]`
- **Sequence of Dequeued Values**: `20, 30, 4, 50, 6, 2, 100, 1`

4(b) Stack Operations (10 Marks) —

Problem Statement

Show the stack after each operation of the following series of stack operations, if executed upon an initially empty stack:

```
push(10), push(20), pop(), push(0), push(1), pop(), pop(),
push(2), push(30), pop(),
push(3), push(4), pop(), pop(), push(100), pop(), pop()
```

Behavioral Rules of Stack

- Last-In, First-Out (LIFO). Elements are pushed and popped strictly at the **top**.
- Mental Model: Like the nested call stack in recursive DNA sequence parsing or RNA secondary structure stem-loop matching.

Step-by-Step Trace

Step	Operation	Stack State (Bottom → Top)	Top Element	Value Returned by [illegible]
1	`push(10)`	`[10]`	10	—
2	`push(20)`	`[10, 20]`	20	—
3	`pop()`	`[10]`	10	**20**
4	`push(0)`	`[10, 0]`	0	—
5	`push(1)`	`[10, 0, 1]`	1	—
6	`pop()`	`[10, 0]`	0	**1**
7	`pop()`	`[10]`	10	**0**
8	`push(2)`	`[10, 2]`	2	—
9	`push(30)`	`[10, 2, 30]`	30	—
10	`pop()`	`[10, 2]`	2	**30**
11	`push(3)`	`[10, 2, 3]`	3	—
12	`push(4)`	`[10, 2, 3, 4]`	4	—
13	`pop()`	`[10, 2, 3]`	3	**4**
14	`pop()`	`[10, 2]`	2	**3**
15	`push(100)`	`[10, 2, 100]`	100	—
16	`pop()`	`[10, 2]`	2	**100**
17	`pop()`	`[10]`	10	**2**

- **Final Stack State**: `[10]` (Top is `10`)
- **Sequence of Popped Values**: `20, 1, 0, 30, 4, 3, 100, 2`

Question 5: Queue Reversal (10 Marks)

Problem Statement — Given an integer k and a queue of integers, implement a function to reverse the order of the first k elements of the queue while keeping the other elements in the same original order.
Allowed Queue Operations Only: - `enqueue(x)` : Add item x to rear of queue. - `dequeue()` : Remove item from front of queue. - `size()` : Returns number of elements. - `front()` : Returns front item without removing.

Examples

- **Input**: $q = [1, 2, 3, 4, 5], k = 3 \Longrightarrow$ **Output**: $[3, 2, 1, 4, 5]$
- **Input**: $q = [4, 3, 2, 1], k = 4 \Longrightarrow$ **Output**: $[1, 2, 3, 4]$

Algorithm Design & The LIFO Inversion Principle — A Queue is FIFO (First-In, First-Out), which preserves order. A Stack is LIFO (Last-In, First-Out), which **naturally reverses order** (just like taking the reverse complement of an oligonucleotide).
To reverse the first k elements while preserving the remaining $n - k$ elements: 1. **Extract & Reverse**: Dequeue the first k elements from the queue and push them onto an auxiliary Stack. 2. **Re-insert Reversed**: Pop all k elements from the stack and enqueue them back into the queue. They are now at the *back* of the queue in reversed order. 3. **Rotate Remaining Elements**: The remaining $n - k$ elements are currently sitting at the front of the queue in their original order. Dequeue and immediately enqueue them one-by-one ($n - k$ times). This rotates them behind the reversed prefix, restoring their original relative order.

```
flowchart TD
    subgraph Step1["Step 1: Dequeue k items to Stack"]
        Q1["Queue: [4, 5]"]
        S1["Stack: [1, 2, 3] (3 on top)"]
    end
    subgraph Step2["Step 2: Pop Stack back into Queue"]
        Q2["Queue: [4, 5, 3, 2, 1]"]
    end
    subgraph Step3["Step 3: Rotate (size - k) elements to back"]
        Q3["Queue: [3, 2, 1, 4, 5]"]
    end
    Step1 --> Step2 --> Step3
```

Complete Java Implementation —

```
import java.util.Queue;
import java.util.Stack;

public class QueueReversal {

    public static void reverseFirstK(Queue<Integer> q, int k) {
        // Validation / Edge Cases
        if (q == null || k <= 0 || k > q.size()) {
            return;
        }

        Stack<Integer> stack = new Stack<>();

        // Step 1: Dequeue first k elements and push onto stack
        for (int i = 0; i < k; i++) {
            stack.push(q.poll()); // dequeue
        }

        // Step 2: Pop from stack and enqueue back to the queue
(reversed order)
        while (!stack.isEmpty()) {
            q.add(stack.pop()); // enqueue
        }

        // Step 3: Rotate the remaining (size - k) elements to the
rear
        int remaining = q.size() - k;
        for (int i = 0; i < remaining; i++) {
            q.add(q.poll()); // dequeue front and enqueue back
        }
    }
}
```

Complexity Analysis —
- **Time Complexity**:
  – Pushing k elements to stack: $\mathcal{O}(k)$
  – Popping k elements to queue: $\mathcal{O}(k)$
  – Rotating $(n - k)$ elements: $\mathcal{O}(n - k)$
  – Total Time: $\mathcal{O}(k + k + n - k) = \mathcal{O}(n)$, strictly linear.
- **Space Complexity**: $\mathcal{O}(k)$ auxiliary space for the auxiliary stack.

SECTION 12: FULL MOCK EXAM 2 WALKTHROUGH & PROOFS

Mock Exam 2: Detailed Solutions & Explanations

Question 1: Sorting (20 Marks)

1(a) Concept of Stability in Sorting (10 Marks) —

Definition of Stability

A sorting algorithm is defined as **stable** if and only if elements with equal keys maintain their original relative order (their input sequence) in the sorted output.

```
flowchart LR
    subgraph Input["Input Array"]
        A1["Card(5, Hearts) [Index 0]"]
        A2["Card(3, Spades) [Index 1]"]
        A3["Card(5, Diamonds) [Index 2]"]
    end
    subgraph Stable["Stable Output: 5(Hearts) before
5(Diamonds)"]
        B1["Card(3, Spades)"] --> B2["Card(5, Hearts)"] -->
B3["Card(5, Diamonds)"]
    end
    subgraph Unstable["Unstable Output: Order Inverted!"]
        C1["Card(3, Spades)"] --> C2["Card(5, Diamonds)"] -->
C3["Card(5, Hearts)"]
    end
    Input --> Stable
    Input --> Unstable
```

Bioinformatics Mental Model

In computational genomics, multi-pass sorting is standard practice. Suppose you have a table of genetic variations (SNPs) sorted by chromosome position (`POS`). If you subsequently sort by chromosome identifier (`CHROM`) using a **stable** sort, all SNPs on chromosome 1 will remain sorted by position. If the sort is **unstable**, SNPs on chromosome 1 will be scrambled into random coordinate order, breaking downstream binary searches and coordinate indexing.

Stability Analysis: Insertion Sort vs. Selection Sort

1. **Insertion Sort is STABLE**:
   - **Why**: When inserting an element `key`, Insertion Sort scans backwards from right to left through the sorted subarray. It only shifts an element $A[j]$ if $A[j] > \text{key}$ (strictly greater).
   - If an element equal to `key` is encountered ($A[j] == \text{key}$), the strict inequality fails, the scan stops, and the key is placed immediately to its right.
   - Therefore, elements with identical values never cross or jump over each other. Their relative input order is preserved.
2. **Selection Sort is UNSTABLE**:
   - **Why**: Selection Sort finds the extremum (minimum or maximum) and swaps it with the element at the boundary of the unsorted subarray. This long-range swap can leapfrog over identical elements located in between.
   - **Concrete Counterexample**: Consider sorting the 3-element array:
     $$[2_a, 2_b, 1]$$
     (where 2_a and 2_b have equal key value 2, but 2_a appears first).
     – Iteration 1: The minimum element in the array is 1 at index 2. Selection Sort swaps 1 with the element at index 0 (2_a).
     – The array becomes:
     $$[1, 2_b, 2_a]$$
     – Result: 2_b is now placed *before* 2_a. Their relative order is inverted. Hence, Selection Sort is fundamentally unstable.

1(b) Best-Case Comparison: Insertion Sort vs. Selection Sort (10 Marks) —

Assume an array of size n that is **already sorted in ascending order** (Best Case).

Metric	Insertion Sort (Best Case)	Selection Sort (Best Case)	Key Conceptual Difference
Comparisons	$n - 1$ ($\mathcal{O}(n)$)	$\frac{n(n-1)}{2}$ ($\mathcal{O}(n^2)$)	Insertion Sort exits early; Selection Sort blindly scans everything.
Swaps / Shifts	0 shifts ($\mathcal{O}(1)$)	0 swaps (or $n - 1$ self-swaps)	Neither algorithm moves data in the already-sorted case.
Time Complexity	$\mathcal{O}(n)$ (Linear)	$\mathcal{O}(n^2)$ (Quadratic)	Insertion Sort is adaptive; Selection Sort is completely non-adaptive.

Detailed Mechanical Explanation

1. **Insertion Sort**:
   - For each element from index $i = 1$ to $n - 1$, the key is compared to its immediate predecessor $A[i - 1]$.
   - Because the array is already sorted, $A[i - 1] \leq \text{key}$ is immediately true on the very first comparison.
   - The inner while-loop never executes any shifts and terminates after exactly **1 comparison per outer loop step**.
   - Total comparisons: $\sum_{i=1}^{n-1} 1 = n - 1$.
   - Total data movements: 0 elements shifted.
   - Overall Best-Case Time: $\mathcal{O}(n)$.
2. **Selection Sort**:
   - To place the minimum element at index i, Selection Sort has no mechanism to know that the subarray $A[i...n - 1]$ is already sorted.
   - It is forced to scan every single element from $i + 1$ to $n - 1$ to verify that no smaller element exists.
   - For $i = 0$, it makes $n - 1$ comparisons; for $i = 1$, $n - 2$ comparisons, all the way to 1.
   - Total comparisons:
     $$\sum_{i=0}^{n-2}(n - 1 - i) = (n - 1) + (n - 2) + ... + 1 = \frac{n(n-1)}{2} \in \mathcal{O}(n^2)$$
   - Total swaps: With an optimization check (`if (min_idx != i)`), 0 swaps are performed; without it, $n - 1$ redundant self-swaps occur.
   - Overall Best-Case Time: Still $\mathcal{O}(n^2)$.

Question 2: Sorting (15 Marks)

Problem Statement — Given the sequence S:

$$45, 12, 85, 32, 89, 45, 21$$

Provide the detailed steps to arrange S in **ascending order** using the Selection Sort algorithm.
Constraint: In each iteration, you **MUST** find the **MAXIMUM** element and swap it with the element at the **END** of the unsorted portion (instead of the traditional way of finding the minimum and swapping it to the front). Show the changes in S at each iteration.

Step-by-Step Execution —
- Array length: $n = 7$.
- Indices: `0` to `6`.
- Unsorted portion at iteration k is $A[0 \ldots \text{end}]$, where end ranges from 6 down to 1.
- Tie-breaking convention for equal keys: We maintain the standard strict comparison (`if (A[j] > maxVal)`), keeping the first occurrence encountered.

Iteration	Unsorted Range	Max Found (Value @ Index)	End Index	Swap Action	Array State After Iteration
Initial	`A[0..6]`	—	—	—	[45, 12, 85, 32, 89, 45, 21]
1	`A[0..6]`	**89** at index 4	6	Swap $A[4]$ (89) with $A[6]$ (21)	[45, 12, 85, 32, 21, 45, 89]
2	`A[0..5]`	**85** at index 2	5	Swap $A[2]$ (85) with $A[5]$ (45)	[45, 12, 45, 32, 21, 85, 89]
3	`A[0..4]`	**45** at index 0	4	Swap $A[0]$ (45) with $A[4]$ (21)	[21, 12, 45, 32, 45, 85, 89]
4	`A[0..3]`	**45** at index 2	3	Swap $A[2]$ (45) with $A[3]$ (32)	[21, 12, 32, 45, 45, 85, 89]
5	`A[0..2]`	**32** at index 2	2	Max is already at index 2 (self-swap)	[21, 12, 32, 45, 45, 85, 89]
6	`A[0..1]`	**21** at index 0	1	Swap $A[0]$ (21) with $A[1]$ (12)	[12, 21, 32, 45, 45, 85, 89]

[!NOTE] Observe Iteration 3: The element `45` at index 0 was swapped with `21` at index 4, jumping past the second `45` at index 2! This vividly demonstrates in practice why Selection Sort is unstable.

Final Result —

$$[\mathbf{12}, \mathbf{21}, \mathbf{32}, \mathbf{45}, \mathbf{45}, \mathbf{85}, \mathbf{89}]$$

Question 3: Linked List (35 Marks)

Problem Statement — Given the sorted (ascending order) `SinglyLinkList` class below:

```
public class SinglyLinkList {
    private Node first;

    public void moveLastToFront() {
        // YOUR CODE
    }

    public void removeDuplicates() {
        // YOUR CODE
    }
}

public class Node {
    private int key;
    private Node next;
}
```

- **a)** (15 marks) Complete the `moveLastToFront()` method that removes the last node of the list and makes it the new first node. (You must handle edge cases like an empty list or a list with only one node).
- **b)** (20 marks) Complete the `removeDuplicates()` method that removes all duplicate values from the sorted list, keeping only the first occurrence of each value.

Solution 3(a): `moveLastToFront()` —

Edge Cases & Pointer Manipulation

1. **Edge Case (0 or 1 Node)**: If `first == null` (empty list) or `first.next == null` (single node), moving the last to front changes nothing. Return immediately.
2. **General Case (≥ 2 Nodes)**:
   - We need to locate the **second-to-last node** (`secLast`), because its `next` pointer must be set to `null` to detach the last node cleanly.
   - Traverse: `while (secLast.next.next != null) secLast = secLast.next;`.
   - Identify `last = secLast.next`.
   - Sever link: `secLast.next = null;`.
   - Prepend: `last.next = first;`.
   - Update head: `first = last;`.

```
flowchart LR
    subgraph Before["Before: 10 -> 20 -> 30 -> 40 -> null"]
        N1["10"] --> N2["20"] --> N3["30 (secLast)"] --> N4["40 (last)"]
    end
    subgraph After["After: 40 -> 10 -> 20 -> 30 -> null"]
        M4["40 (first)"] --> M1["10"] --> M2["20"] --> M3["30 (next = null)"]
    end
```

Java Implementation

```
public void moveLastToFront() {
    // Edge case: Empty list or list with only one node
    if (first == null || first.next == null) {
        return;
    }

    // Traverse to find the second-to-last node
    Node secLast = first;
    while (secLast.next.next != null) {
        secLast = secLast.next;
    }

    // secLast.next is the last node
    Node last = secLast.next;

    // Disconnect the last node from the end
    secLast.next = null;

    // Attach last node before current first node
    last.next = first;

    // Make last node the new head
    first = last;
}
```

- **Time Complexity**: $\mathcal{O}(n)$ because we must traverse the list once to reach the second-to-last node.
- **Space Complexity**: $\mathcal{O}(1)$ auxiliary space.

Solution 3(b): `removeDuplicates()` —

Algorithm & Pointer Manipulation

Because the linked list is **already sorted in ascending order**, all identical values are contiguous (adjacent in memory). 1. If the list is empty (`first == null`), do nothing. 2. Maintain a pointer `current = first`. 3. While `current.next != null`: - If `current.key == current.next.key`, bypass the duplicate node: `java current.next = current.next.next;` *(Crucial: Do not advance `current` here, because the newly spliced `current.next` might also be a duplicate, e.g., in `1 -> 1 -> 1`).* - If `current.key != current.next.key`, advance: `java current = current.next;`

Java Implementation

```
public void removeDuplicates() {
    if (first == null) {
        return;
    }

    Node current = first;
    while (current.next != null) {
        if (current.key == current.next.key) {
            // Duplicate detected: bypass current.next
            current.next = current.next.next;
        } else {
            // Distinct value: advance to the next node
            current = current.next;
        }
    }
}
```

- **Time Complexity**: $\mathcal{O}(n)$ single linear pass.
- **Space Complexity**: $\mathcal{O}(1)$ in-place modification.

Question 4: Queue and Stack (20 Marks)

4(a) Min Priority Queue Operations (10 Marks) —

Problem Statement

Consider a **MIN PRIORITY QUEUE**. Execute the following sequence of operations on an initially empty queue and write down the exact sequence of values returned by the `dequeue()` operations:

```
enqueue(15), enqueue(10), enqueue(20), dequeue(), enqueue(5), dequeue(),
dequeue(), enqueue(30), enqueue(2), dequeue()
```

Behavioral Rules

- `enqueue(x)`: Adds item x.
- `dequeue()`: Always removes and returns the **minimum numerical value** currently present in the queue.

Step-by-Step Trace

Step	Operation	Priority Queue Contents (Sorted Ascending)	Value Returned by `dequeue()`
1	`enqueue(15)`	`[15]`	—
2	`enqueue(10)`	`[10, 15]`	—
3	`enqueue(20)`	`[10, 15, 20]`	—
4	`dequeue()`	Remove min (10) ⟹ Queue: `[15, 20]`	**10**
5	`enqueue(5)`	`[5, 15, 20]`	—
6	`dequeue()`	Remove min (5) ⟹ Queue: `[15, 20]`	**5**
7	`dequeue()`	Remove min (15) ⟹ Queue: `[20]`	**15**
8	`enqueue(30)`	`[20, 30]`	—
9	`enqueue(2)`	`[2, 20, 30]`	—
10	`dequeue()`	Remove min (2) ⟹ Queue: `[20, 30]`	**2**

Final Answer

The exact sequence of values returned by the `dequeue()` operations is:

$$\mathbf{10}, \mathbf{5}, \mathbf{15}, \mathbf{2}$$

4(b) Circular Queue Trace (10 Marks) —

Problem Statement

Consider a Circular Queue implemented using an array A of maximum capacity $N = 5$ (indices 0 to 4). Initially:

$$\text{front} = 0, \quad \text{rear} = -1, \quad \text{size} = 0$$

Show the exact contents of the array A, and the values of the `front` and `rear` indices after executing the following sequence:

```
enqueue(10), enqueue(20), enqueue(30), dequeue(), enqueue(40),
enqueue(50), enqueue(60), dequeue()
```

Formulas for Array-Based Circular Queue

- enqueue(x): $\text{rear} = (\text{rear} + 1) \pmod N$; $A[\text{rear}] = x$; $\text{size} = \text{size} + 1$.
- dequeue(): $x = A[\text{front}]$; $\text{front} = (\text{front} + 1) \pmod N$; $\text{size} = \text{size} - 1$.

Step-by-Step Trace

Step	Operation	front	rear	size	Array A Contents [0, 1, 2, 3, 4]	Explanation
0	Initial	0	−1	0	[null, null, null, null, null]	Empty queue
1	`enqueue(10)`	0	0	1	[10, null, null, null, null]	rear = $(-1+1) \pmod 5 = 0$; store 10
2	`enqueue(20)`	0	1	2	[10, 20, null, null, null]	rear = $(0+1) \pmod 5 = 1$; store 20
3	`enqueue(30)`	0	2	3	[10, 20, 30, null, null]	rear = $(1+1) \pmod 5 = 2$; store 30
4	`dequeue()`	1	2	2	[10, 20, 30, null, null]	Value 10 dequeued; front = $(0+1) \pmod 5 = 1$
5	`enqueue(40)`	1	3	3	[10, 20, 30, 40, null]	rear = $(2+1) \pmod 5 = 3$; store 40
6	`enqueue(50)`	1	4	4	[10, 20, 30, 40, 50]	rear = $(3+1) \pmod 5 = 4$; store 50
7	`enqueue(60)`	1	**0**	5	[60, 20, 30, 40, 50]	rear **wraps around**: $(4+1) \pmod 5 = 0$; 60 overwrites index 0
8	`dequeue()`	**2**	**0**	4	[60, 20, 30, 40, 50]	Value 20 dequeued; front = $(1+1) \pmod 5 = 2$

Final State

- **Array A Contents**: `[60, 20, 30, 40, 50]` *(In memory, index 1 still contains 20 unless cleared, but logically the queue contains indices 2, 3, 4, 0).*
- **Active Elements (Front to Rear)**: `30, 40, 50, 60`
- **front Index**: `2`
- **rear Index**: `0`
- **size**: `4`

Question 5: Stack & Queue (10 Marks)

Problem Statement — Given a stack of integers, implement a function to sort the stack in **ascending order** (smallest items on top).
Constraint: You are allowed to use **exactly one additional auxiliary Stack**. You may **NOT** use arrays, queues, lists, or any other data structures.
Only the following operations are allowed on a stack: - `push(x)`: Add an item x to the top. - `pop()`: Remove and return the top item. - `peek()`: Return the top item without removing it. - `isEmpty()`: Check if the stack is empty.

Example

- **Input Stack (Top is right)**: `[34, 3, 31, 98, 92, 23]` (23 on top)
- **Output Stack (Top is right)**: `[98, 92, 34, 31, 23, 3]` (3 on top)

Algorithm Walkthrough — Let the given stack be `input` and our auxiliary stack be `aux`.

The Inversion Principle

If we want the final `input` stack to have the **smallest items on top** (ascending from top down, meaning largest at the bottom), our intermediate `aux` stack must hold elements in **descending order** (largest items on top). When we finally pour `aux` back into `input`, the order inverts, placing the smallest items on top!

Step-by-Step Procedure

1. While `input` is not empty:

- Pop the top element from `input` and store it in a temporary variable: `tmp = input.pop()`.
- While `aux` is not empty AND `aux.peek() > tmp`:
  - Pop from `aux` and push it back into `input`: `input.push(aux.pop())`. *(This shifts larger elements out of the way, transferring them temporarily back to `input`).*
- Push `tmp` onto `aux`.

2. Once `input` is empty, all elements are in `aux` sorted with the largest element at the top.
3. Transfer all elements from `aux` back to `input` (`input.push(aux.pop())`).
4. `input` now contains all elements sorted with the **smallest item on top**.

```
flowchart TD
    A["input has elements?"] -- Yes --> B["tmp = input.pop()"]
    B --> C{"aux not empty AND<br/>aux.peek() > tmp?"}
    C -- Yes --> D["input.push(aux.pop())"]
    D --> C
    C -- No --> E["aux.push(tmp)"]
    E --> A
    A -- No --> F["Pour aux back to input<br/
>input.push(aux.pop())"]
    F --> G["Done! Smallest on top."]
```

Complete Java Implementation —

```
import java.util.Stack;

public class StackSorter {

    public static void sortStack(Stack<Integer> input) {
        if (input == null || input.isEmpty()) {
            return;
        }

        Stack<Integer> aux = new Stack<>();

        // Process each element from the input stack
        while (!input.isEmpty()) {
            int tmp = input.pop();

            // Shift elements in aux that are greater than tmp back
into input
            while (!aux.isEmpty() && aux.peek() > tmp) {
                input.push(aux.pop());
            }

            // Place tmp in its correct sorted position in aux
            aux.push(tmp);
        }

        // Transfer elements back to input stack:
        // aux has largest on top -> pouring into input puts
smallest on top!
        while (!aux.isEmpty()) {
            input.push(aux.pop());
        }
    }
}
```

Complexity Analysis —

- **Time Complexity**: $\mathcal{O}(n^2)$ in the worst case (e.g., if the input stack is already sorted in reverse order, each insertion shifts elements back and forth between the two stacks, identical to Insertion Sort mechanics).
- **Space Complexity**: $\mathcal{O}(n)$ auxiliary space to hold the n elements in the single additional `aux` stack.

SECTION 13: FULL REAL EXAM 2023 WALKTHROUGH & PROOFS

Real Exam 2023 (IT013IU HK1 Midterm): Complete Solutions & Explanations

---

Exam Information

- **Course**: Algorithms & Data Structures (IT013IU)
- **Institution**: International University – VNU-HCM (School of Computer Science and Engineering)
- **Exam Date**: 03/04/2023
- **Duration**: 90 minutes
- **Total Marks**: 100 marks (equivalent to 30% of course grade)

Question 1: Sorting (20 Marks)

Problem Statement — Given array A:

Index	0	1	2	3	4	5	6	7	8	9	10	11	12
Data	72	4	12	53	15	2	8	16	40	21	7	87	35

Sort the array with **Selection Sort** by filling the table below.

Mechanics of Selection Sort — Selection Sort operates by progressively expanding a sorted prefix on the left: 1. For each position i from 0 to $n-2$ (here indices 0 to 11): - Find the index `min_idx` of the minimum element in the remaining unsorted subarray $A[i...12]$. - Swap $A[i]$ with $A[\text{min_idx}]$. 2. After iteration i, the element at index i is guaranteed to be in its final sorted position.

Step-by-Step Selection Sort Execution Table —

Iteration / Action	0	1	2	3	4	5	6	7	8	9	10	11	12
Initial Array	72	4	12	53	15	2	8	16	40	21	7	87	35
Step 1 (Min = 2 @ idx 5 ↔ Swap $A[0]$)	**2**	4	12	53	15	72	8	16	40	21	7	87	35
Step 2 (Min = 4 @ idx 1 ↔ Self-swap $A[1]$)	**2**	**4**	12	53	15	72	8	16	40	21	7	87	35
Step 3 (Min = 7 @ idx 10 ↔ Swap $A[2]$)	**2**	**4**	**7**	53	15	72	8	16	40	21	12	87	35
Step 4 (Min = 8 @ idx 6 ↔ Swap $A[3]$)	**2**	**4**	**7**	**8**	15	72	53	16	40	21	12	87	35
Step 5 (Min = 12 @ idx 10 ↔ Swap $A[4]$)	**2**	**4**	**7**	**8**	**12**	72	53	16	40	21	15	87	35
Step 6 (Min = 15 @ idx 10 ↔ Swap $A[5]$)	**2**	**4**	**7**	**8**	**12**	**15**	53	16	40	21	72	87	35
Step 7 (Min = 16 @ idx 7 ↔ Swap $A[6]$)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	53	40	21	72	87	35
Step 8 (Min = 21 @ idx 9 ↔ Swap $A[7]$)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	40	53	72	87	35
Step 9 (Min = 35 @ idx 12 ↔ Swap $A[8]$)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	53	72	87	40
Step 10 (Min = 40 @ idx 12 ↔ Swap $A[9]$)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	**40**	72	87	53
Step 11 (Min = 53 @ idx 12 ↔ Swap $A[10]$)	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	**40**	**53**	87	72
Step 12	**2**	**4**	**7**	**8**	**12**	**15**	**16**	**21**	**35**	**40**	**53**	**72**	**87**

Final Sorted Array

$$[2, 4, 7, 8, 12, 15, 16, 21, 35, 40, 53, 72, 87]$$

Question 2: Queue and Stack (20 Marks)

2(a) FIFO Queue Trace (10 Marks) —

Problem Statement

What values are returned after each `dequeue()` during the following sequence of queue operations, if executed on an initially empty queue?

```
enqueue(0), enqueue(3), dequeue(), enqueue(0), enqueue(4),
dequeue(), dequeue(),
enqueue(2), enqueue(0), dequeue(), enqueue(2), enqueue(3),
dequeue(), dequeue(),
enqueue(1330), dequeue(), dequeue()
```

Behavioral Rule

Standard FIFO (First-In, First-Out). Elements enter at the back (rear) and exit at the front.

Step-by-Step Trace

Step	Operation	Queue State (Front → Rear)	Returned by [redacted]
1	`enqueue(0)`	`[0]`	—
2	`enqueue(3)`	`[0, 3]`	—
3	`dequeue()`	`[3]`	**0**
4	`enqueue(0)`	`[3, 0]`	—
5	`enqueue(4)`	`[3, 0, 4]`	—
6	`dequeue()`	`[0, 4]`	**3**
7	`dequeue()`	`[4]`	**0**
8	`enqueue(2)`	`[4, 2]`	—
9	`enqueue(0)`	`[4, 2, 0]`	—
10	`dequeue()`	`[2, 0]`	**4**
11	`enqueue(2)`	`[2, 0, 2]`	—
12	`enqueue(3)`	`[2, 0, 2, 3]`	—
13	`dequeue()`	`[0, 2, 3]`	**2**
14	`dequeue()`	`[2, 3]`	**0**
15	`enqueue(1330)`	`[2, 3, 1330]`	—
16	`dequeue()`	`[3, 1330]`	**2**
17	`dequeue()`	`[1330]`	**3**

Final Sequence of Values Returned by `dequeue()`

$$\mathbf{0, 3, 0, 4, 2, 0, 2, 3}$$

(Notice the hidden easter egg: `03, 04, 2023` encodes the exact date of the examination: April 3, 2023!)

2(b) LIFO Stack Trace (10 Marks) —

Problem Statement

What values are returned after each `pop()` during the following series of stack operations, if executed upon an initially empty stack?

```
push(0), push(3), pop(), push(0), push(4), pop(), pop(), push(2),
push(0), pop(),
push(2), push(3), pop(), pop(), push(1330), pop(), pop()
```

Behavioral Rule

Standard LIFO (Last-In, First-Out). Elements are added and removed from the **top** of the stack.

Step-by-Step Trace

Step	Operation	Stack State (Bottom → Top)	Top Element	Returned by [redacted]
1	`push(0)`	`[0]`	0	—
2	`push(3)`	`[0, 3]`	3	—
3	`pop()`	`[0]`	0	**3**
4	`push(0)`	`[0, 0]`	0	—
5	`push(4)`	`[0, 0, 4]`	4	—
6	`pop()`	`[0, 0]`	0	**4**
7	`pop()`	`[0]`	0	**0**
8	`push(2)`	`[0, 2]`	2	—
9	`push(0)`	`[0, 2, 0]`	0	—
10	`pop()`	`[0, 2]`	2	**0**
11	`push(2)`	`[0, 2, 2]`	2	—
12	`push(3)`	`[0, 2, 2, 3]`	3	—
13	`pop()`	`[0, 2, 2]`	2	**3**
14	`pop()`	`[0, 2]`	2	**2**
15	`push(1330)`	`[0, 2, 1330]`	1330	—
16	`pop()`	`[0, 2]`	2	**1330**
17	`pop()`	`[0]`	0	**2**

Final Sequence of Values Returned by `pop()`

$$3, 4, 0, 0, 3, 2, 1330, 2$$

Question 3: Linked List (30 Marks)

Problem Statement — Given the 2 classes `Node` and `DoublyLinkList`:

```
public class DoublyLinkList {
    public Node first;
    public Node last;
}

public class Node {
    public int key;
    public Node next;
    public Node previous;
}
```

- **(5 marks)** Implement function `delete_first` for the doubly linked list.
- **(5 marks)** Implement function `insert_last` for the doubly linked list.
- **(20 marks)** Implement an algorithm to merge two sorted doubly linked lists to make a new sorted linked list.

3.1 Function `delete_first()` (5 Marks) —

Logic & Edge Cases

- **Empty List**: If `first == null`, nothing to delete.
- **Single Node** (`first == last`): Set both `first = null` and `last = null`.
- **Multiple Nodes**: Advance `first = first.next`, and sever the reverse link by setting `first.previous = null`.

```
public void delete_first() {
    if (first == null) {
        return; // Empty list: do nothing
    }
    if (first == last) {
        // Only one node in the list
        first = null;
        last = null;
    } else {
        // Multiple nodes: move head forward and detach back-link
        first = first.next;
        first.previous = null;
    }
}
```

- **Time Complexity**: $\mathcal{O}(1)$ constant time.
- **Space Complexity**: $\mathcal{O}(1)$.

3.2 Function `insert_last(int key)` (5 Marks) —

Logic & Edge Cases

- Create a `newNode` with `newNode.key = key`.
- **Empty List** (`first == null`): Both `first` and `last` point to `newNode`.
- **Non-Empty List**:
  - `last.next = newNode;`
  - `newNode.previous = last;`
  - `last = newNode;`

```
public void insert_last(int key) {
    Node newNode = new Node();
    newNode.key = key;
    newNode.next = null;
    newNode.previous = null;

    if (first == null) {
        // List was empty
        first = newNode;
        last = newNode;
    } else {
        // Attach to current last and update last pointer
        last.next = newNode;
        newNode.previous = last;
        last = newNode;
    }
}
```

- **Time Complexity**: $\mathcal{O}(1)$ constant time (since a direct `last` pointer is maintained).
- **Space Complexity**: $\mathcal{O}(1)$.

3.3 Algorithm to Merge Two Sorted Doubly Linked Lists (20 Marks) —

Bioinformatics Mental Model

This operation is the exact mechanical foundation of **Merge Sort** and coordinate-based genomic interval merging (like merging two BAM/CRAM files or sorted VCF variant calls). We have two streams of ordered genomic records; we compare their heads and repeatedly pick the smaller one to build an integrated sorted sequence.

Design Decision: Non-Destructive vs. In-Place Splicing

The problem asks to *"make a new sorted linked list"*. Creating a clean `result` list and appending via `insert_last` preserves the integrity of the original lists and eliminates brittle pointer overwriting bugs.

Java Implementation

```
public static DoublyLinkList merge(DoublyLinkList list1,
DoublyLinkList list2) {
    DoublyLinkList result = new DoublyLinkList();

    // Guard against null list objects
    if (list1 == null && list2 == null) return result;
    if (list1 == null || list1.first == null) return list2;
    if (list2 == null || list2.first == null) return list1;

    Node p1 = list1.first;
    Node p2 = list2.first;

    // Two-pointer merge: compare current nodes from both lists
    while (p1 != null && p2 != null) {
        if (p1.key <= p2.key) {
            result.insert_last(p1.key);
            p1 = p1.next;
        } else {
            result.insert_last(p2.key);
            p2 = p2.next;
        }
    }

    // Append any remaining elements from list1
    while (p1 != null) {
        result.insert_last(p1.key);
        p1 = p1.next;
    }

    // Append any remaining elements from list2
    while (p2 != null) {
        result.insert_last(p2.key);
        p2 = p2.next;
    }

    return result;
}
```

- **Time Complexity**: $\mathcal{O}(n_1 + n_2)$, strictly linear in the total number of nodes. Each node is inspected once.
- **Space Complexity**: $\mathcal{O}(n_1 + n_2)$ for the newly created merged list.

Question 4: Complexity (20 Marks)

Problem Statement — Propose the **worst-case complexity** of the following operations in linked-lists:

Operation	Unsorted, simple (singly) linked list	Sorted simple (singly) linked list	Unsorted, double (doubly) linked list	Sorted double (doubly) linked list
Search	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$
Insert a new value	$\mathcal{O}(1)$	$\mathcal{O}(n)$	$\mathcal{O}(1)$	$\mathcal{O}(n)$
Delete a given value	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$
Get minimum	$\mathcal{O}(n)$	$\mathcal{O}(1)$	$\mathcal{O}(n)$	$\mathcal{O}(1)$
Get maximum	$\mathcal{O}(n)$	$\mathcal{O}(n)$ *(or $\mathcal{O}(1)$ with tail)*	$\mathcal{O}(n)$	$\mathcal{O}(1)$

In-Depth Explanations & Common Exam Pitfalls —

1. Search ($\mathcal{O}(n)$ across all four)

- Linked lists are non-contiguous in memory. Unlike arrays, you cannot calculate memory addresses via index arithmetic (`arr[i] = base + i * size`). You cannot jump directly to the middle to perform Binary Search.
- To search for a key, you must follow pointers sequentially from the start.
- Even if the list is sorted, in the worst case (the target element is at the very end or not present at all), you must traverse all n nodes. Hence, all four are $\mathcal{O}(n)$.

2. Insert a new value

- **Unsorted Lists ($\mathcal{O}(1)$)**: Because there is no ordering constraint, a new node can simply be inserted at the head of the list in $\mathcal{O}(1)$ constant time (or at the tail in $\mathcal{O}(1)$ if a tail pointer is maintained).
- **Sorted Lists ($\mathcal{O}(n)$)**: To maintain the sorted invariant, you **must first search** for the correct location where the value belongs. Scanning to the correct slot takes $\mathcal{O}(n)$ in the worst case. Once the position is found, pointer rewiring takes $\mathcal{O}(1)$, but the total insertion time is dominated by the search: $\mathcal{O}(n) + \mathcal{O}(1) = \mathcal{O}(n)$. > [!WARNING] > **Exam Trap Alert**: Students often mistakenly write $\mathcal{O}(1)$ for sorted list insertion thinking only of pointer splicing. But in a sorted list, you cannot splice until you find *where* to splice, which requires an $\mathcal{O}(n)$ linear scan!

3. Delete a given value ($\mathcal{O}(n)$ across all four)

- The prompt specifies *"Delete a given value"*, not *"Delete a node given its pointer"*.
- To delete a given value, the algorithm must first traverse the list to locate the node containing that value: this search step takes $\mathcal{O}(n)$.
- In a singly linked list, you also need to find the predecessor node to bridge over the deleted node.
- In a doubly linked list, once the node is located, unlinking takes $\mathcal{O}(1)$ using `node.previous` and `node.next`. However, finding that node still took $\mathcal{O}(n)$ search steps. Total: $\mathcal{O}(n)$.

4. Get minimum

- **Unsorted Lists ($\mathcal{O}(n)$)**: The minimum could be anywhere in the list; you must examine all n nodes to be certain.
- **Sorted Lists ($\mathcal{O}(1)$)**: In ascending order, the minimum is always guaranteed to be the very first node (`first.key`). Reading it takes $\mathcal{O}(1)$ immediate time.

5. Get maximum

- **Unsorted Lists ($\mathcal{O}(n)$)**: Unpredictable position; requires scanning all n nodes.
- **Sorted Simple (Singly) List**:
  - Without a tail pointer: Must traverse the entire chain of n nodes from `first` to reach the end: $\mathcal{O}(n)$.
  - With a tail pointer: If the class maintains a `tail` pointer, accessing `tail.key` is $\mathcal{O}(1)$.
- **Sorted Double (Doubly) List ($\mathcal{O}(1)$)**: As defined in the exam (`DoublyLinkList` maintains `public Node last;`), the maximum is immediately accessible via `last.key` in $\mathcal{O}(1)$ time.

Question 5: Queue (10 Marks)

Problem Statement — Whereas a stack allows insertion and deletion of elements at only one end, and a queue allows insertion at one end and deletion at the other end, a **deque** (double ended queue, pronounced like "deck") allows insertion and deletion at both ends.
Write **04 (four) $\mathcal{O}(1)$-time procedures** to insert elements into and delete elements from both ends of a deque implemented by an array.

Conceptual Architecture of an Array-Based Deque — We use a circular array `arr[]` of fixed capacity $\mathcal{C}$, with variables: - `front` : Index of the front element (initially -1). - `rear` : Index of the rear element (initially -1). - `size` : Current count of stored elements (initially 0).

```
flowchart LR
    subgraph Deque_Circular_Array["Circular Array Buffer"]
        F["<-- insertFront() / deleteFront()"] -->
FrontIndex["front"]
        RearIndex["rear"] --> R["insertRear() / deleteRear() -->"]
    end
```

Index Movement Rules

- **Moving Forward (Clocks-wise)**: `(idx + 1) % capacity`
- **Moving Backward (Counter-clockwise)**: `(idx - 1 + capacity) % capacity` *(Adding `capacity` before modulo ensures the result is never negative in Java).*

The 4 Required $\mathcal{O}(1)$ Procedures —

```
public class ArrayDeque {
    private int[] arr;
    private int front;
    private int rear;
    private int size;
    private int capacity;

    public ArrayDeque(int capacity) {
        this.capacity = capacity;
        this.arr = new int[capacity];
        this.front = -1;
        this.rear = -1;
        this.size = 0;
    }

    public boolean isFull() {
        return size == capacity;
    }

    public boolean isEmpty() {
        return size == 0;
    }

    // 1. insertFront: Insert element at the front end - O(1)
    public void insertFront(int key) {
        if (isFull()) {
            throw new RuntimeException("Deque Overflow: cannot
insert into full deque");
        }
        if (isEmpty()) {
            front = 0;
            rear = 0;
        } else {
            // Decrement front circularly
            front = (front - 1 + capacity) % capacity;
        }
        arr[front] = key;
        size++;
    }

    // 2. insertRear: Insert element at the rear end - O(1)
    public void insertRear(int key) {
        if (isFull()) {
            throw new RuntimeException("Deque Overflow: cannot
insert into full deque");
        }
        if (isEmpty()) {
            front = 0;
            rear = 0;
        } else {
            // Increment rear circularly
            rear = (rear + 1) % capacity;
        }
        arr[rear] = key;
        size++;
    }

    // 3. deleteFront: Delete and return element from front end -
O(1)
    public int deleteFront() {
        if (isEmpty()) {
            throw new RuntimeException("Deque Underflow: cannot
delete from empty deque");
        }
        int removedValue = arr[front];
        if (front == rear) {
            // Only one element was in the deque; reset to empty
            front = -1;
            rear = -1;
        } else {
            // Increment front circularly
            front = (front + 1) % capacity;
        }
        size--;
        return removedValue;
    }

    // 4. deleteRear: Delete and return element from rear end -
O(1)
    public int deleteRear() {
        if (isEmpty()) {
            throw new RuntimeException("Deque Underflow: cannot
delete from empty deque");
        }
        int removedValue = arr[rear];
        if (front == rear) {
            // Only one element was in the deque; reset to empty
            front = -1;
            rear = -1;
        } else {
            // Decrement rear circularly
            rear = (rear - 1 + capacity) % capacity;
        }
        size--;
        return removedValue;
    }
}
```

Why Each Procedure is Strictly $\mathcal{O}(1)$ —

- No elements are shifted or relocated in memory.
- Index adjustments use single arithmetic instructions (`+`, `-`, `%`).
- Memory access is direct through array index addressing (`arr[front]` or `arr[rear]`).
- Time complexity is unconditionally $\mathcal{O}(1)$ in best, average, and worst cases.